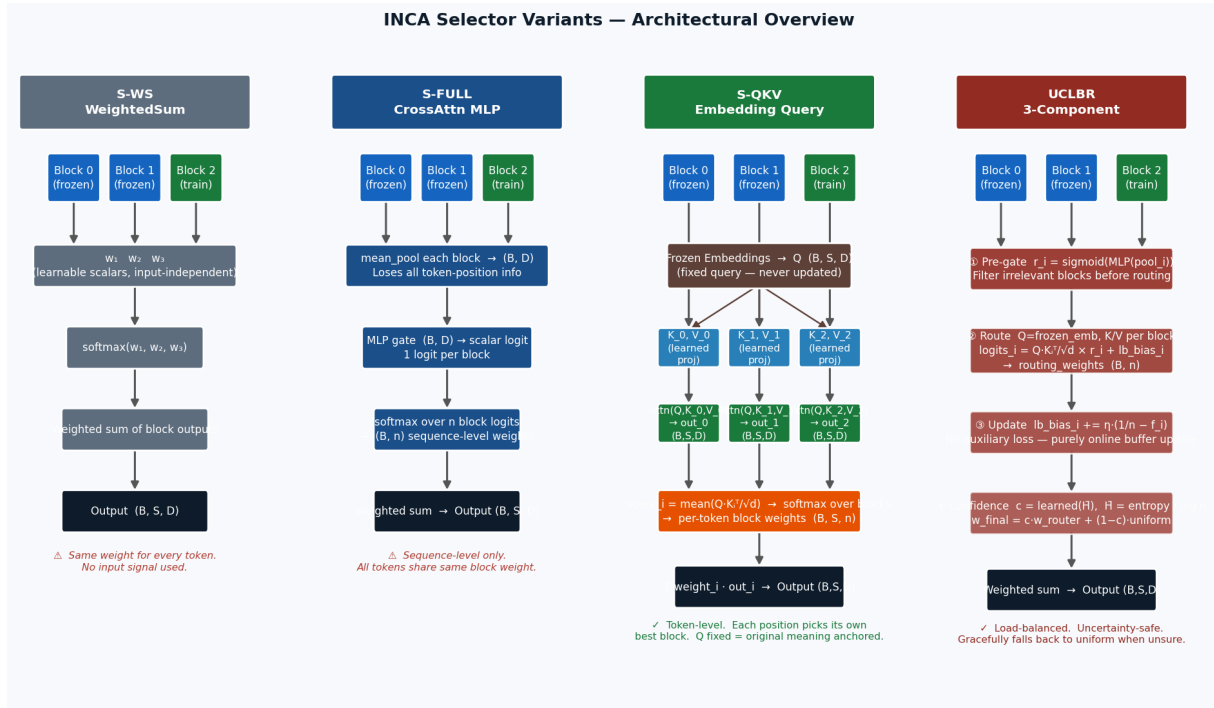


INCA — Selector Architecture

Deep dive: [S-WS](#) · [S-FULL](#) · [S-QKV](#) · [UCLBR](#)

INCA's selector aggregates the outputs of all blocks — frozen and trainable — into a single (B, S, D) tensor that the T5 decoder attends to. Four variants are implemented from Phase 1. All are ablated in experiment E-ROUTE. This section provides full architecture diagrams for each variant and explains the design decisions behind the recommended default (S-QKV).

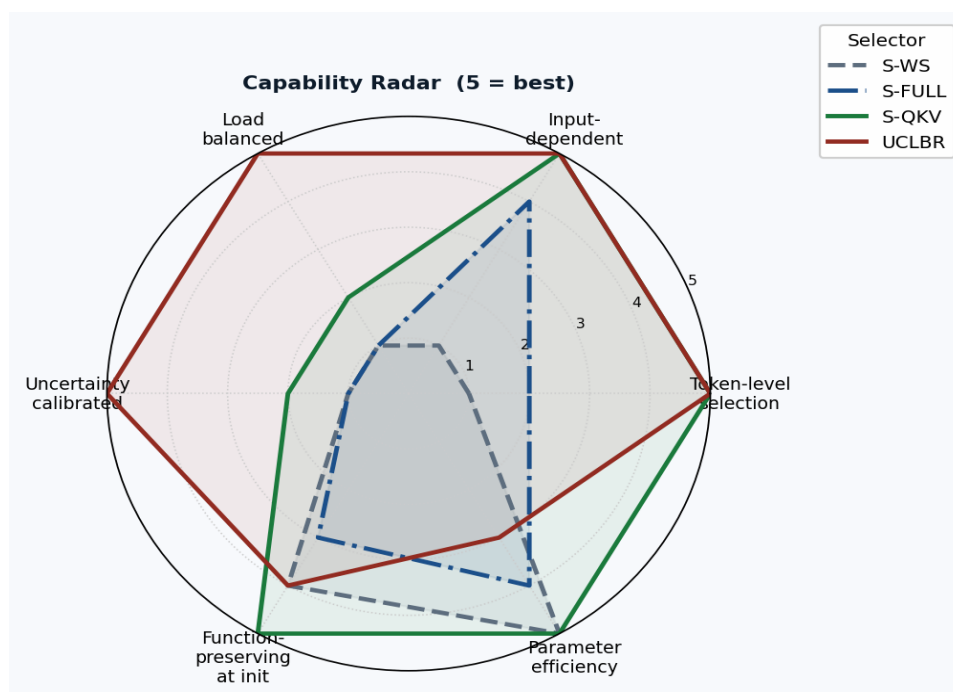
1. Overview — All Four Selectors



Left to right: S-WS (blind scalar weights) · S-FULL (MLP gate, sequence-level) · S-QKV (fixed-query, token-level, per-block separate) · UCLBR (full three-component router)

Selector	YAML key	Granularity	Blocks attended separately?	Recommended use
S-WS	weighted_sum	Sequence	N/A (no attention)	Control ablation only
S-FULL	cross_attention	Sequence	No (mean-pool then MLP)	Ablation baseline
S-QKV	embedding_query	Token-level	Yes ✓	Default for Paper A
UCLBR	uclbr	Token-level	Yes ✓	Extended experiments

2. Capability Radar



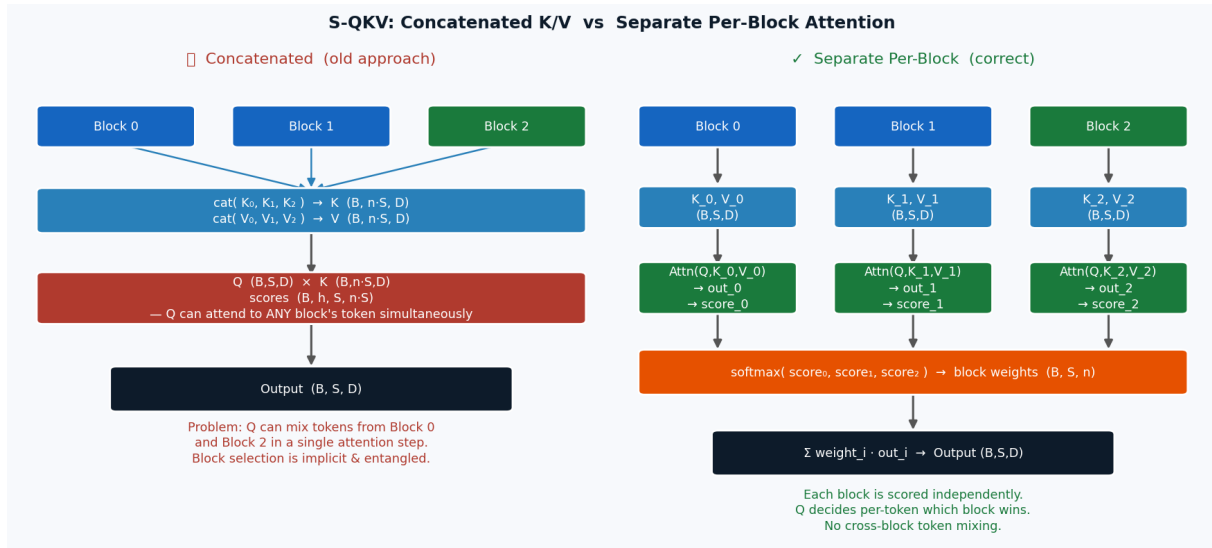
S-QKV and UCLBR both achieve maximum scores on token-level selection and input-dependence. UCLBR adds load-balancing and uncertainty calibration at the cost of implementation complexity and parameter count. S-WS and S-FULL are included as controls — they should be clearly outperformed by S-QKV if token-level routing is genuinely useful.

3. S-QKV — EmbeddingQuerySelector

Core design principle

Q is the frozen embedding — the original meaning of each token before any block has processed it. K and V come from each block's output. The selector asks: "Given what this token originally meant, how well does each block's representation of it serve as an answer?" The block with the highest $Q \cdot K$ alignment wins the most weight.

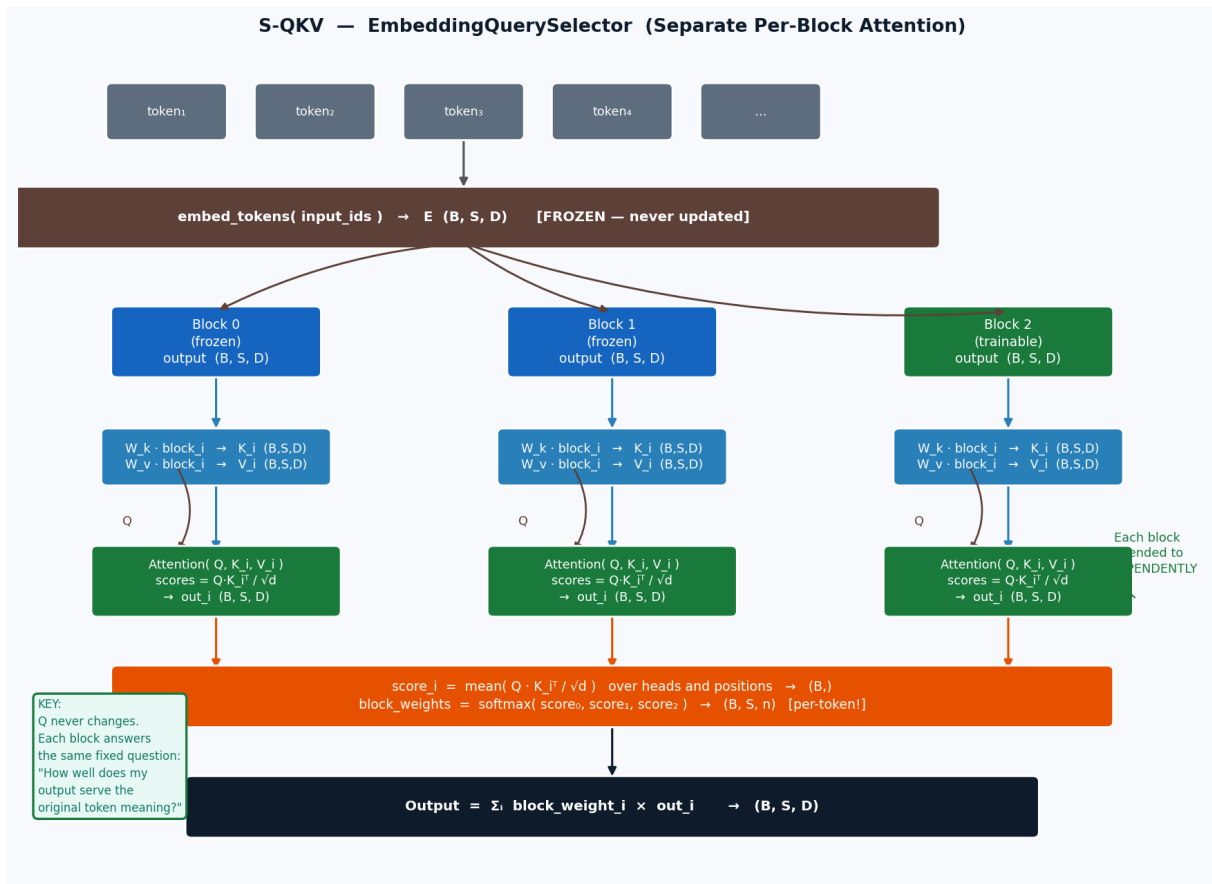
Key design choice: per-block separate attention (not concatenated)



The original (incorrect) implementation concatenated all block K/V tensors into one big matrix: $K = \text{cat}(K_0, K_1, K_2)$. This allowed Q to attend to tokens from Block 0 and Block 2 simultaneously in a single attention step, making block selection implicit and entangled. The correct design attends to each block independently:

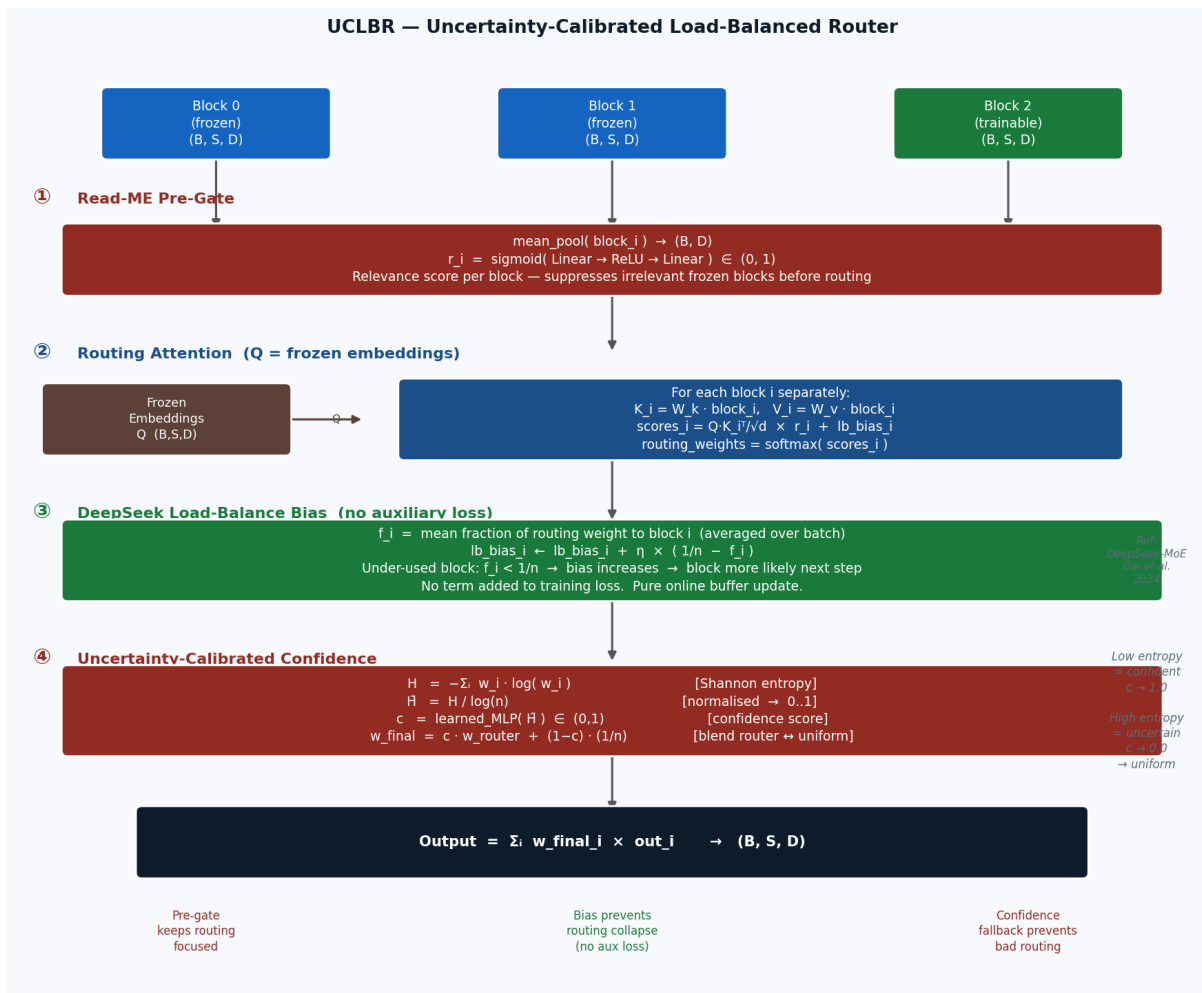
- For each block i : compute $\text{Attention}(Q, K_i, V_i)$ separately $\rightarrow \text{out}_i (B, S, D)$
- Derive a block relevance score from the mean $Q \cdot K_i$ alignment
- Softmax over block scores $\rightarrow$ per-token block weights (B, S, n)
- Final output $= \sum \text{weight}_i \cdot \text{out}_i$

Full data flow



- **Q is never projected.** Raw frozen embeddings are used directly as queries. The question is always the same: what the token originally meant.
- **K and V are identity-initialised.** At grow time the selector is function-preserving — it outputs approximately the mean of all block outputs before any training.
- **Per-token block weights (B, S, n).** Token at position 3 can favour Block 0 while token at position 15 favours Block 2. This is the key advantage over S-FULL.
- **No gradient through Q.** Only W_k , W_v , W_{out} are learnable. The selector learns which blocks serve which kinds of tokens, not what the tokens are.

4. UCLBR — Three-Component Router



Component 1 — Read-ME Pre-gate

Before computing full attention, a two-layer MLP scores each block by mean-pooling its output and computing a sigmoid relevance score $r_i \in (0,1)$. Routing logits are multiplied by r_i before softmax. Frozen blocks that have not yet specialised to the current task are suppressed, preventing their noise from diluting the routing signal of the relevant blocks.

Component 2 — DeepSeek Auxiliary-Loss-Free Load Balancing

Standard MoE load balancing appends an auxiliary loss to the training objective. This can destabilise training by fighting the primary language modelling objective. UCLBR uses the DeepSeek approach: a per-block bias buffer updated purely online after each forward pass — no gradient, no loss term. Under-used blocks accumulate positive bias; over-used blocks are penalised. Load balances naturally.

Component 3 — Uncertainty-Calibrated Confidence

High-entropy routing weights signal that the router is uncertain. When uncertain, committing to a bad routing decision is worse than spreading weight uniformly. The normalised entropy $H = H / \log(n)$ drives a learned confidence score c . The final weights blend the router's decision with a uniform fallback: the more uncertain, the more uniform the output. This prevents catastrophic routing errors during early training or on out-of-distribution

inputs.

5. E-ROUTE — Ablation Protocol

Run all four selectors on CC-News (6 periods, 3 seeds each). Pick the winner by BWT for the TiC-LM main run. The minimum bar: S-QKV must beat S-FULL by $\geq 0.5\%$ BWT absolute, proving token-level routing is worth the complexity.

YAML key	Selector	Hypothesis	Pass condition
weighted_sum	S-WS	Lower bound — input routing adds no value over fixed weights	BWT > S-FULL
cross_attention	S-FULL	Sequence-level gating beats blind weights	BWT > S-WS
embedding_query	S-QKV	Token-level separate-block attention beats sequence-level gating	BWT > S-FULL by $\geq 0.5\%$
uclbr	UCLBR	Load balancing + uncertainty adds further gain over S-QKV	BWT > S-QKV

Decision rule: if $S\text{-QKV} \geq UCLBR$ on BWT, use S-QKV as the Paper A default (simpler story, fewer moving parts). If UCLBR wins by $\geq 1\%$, use UCLBR as default and describe S-QKV as the ablated simplified version. Report both in Table 2.